Extra credit problems

Math 427

0. Find a mistake or misprint in the book. (The score depends on the type of mistake.)

1. Describe all the motions of the Manhattan plane.

2. Construct a metric space $\mathcal{X}$ and a distance-preserving map $f\colon \mathcal{X} \to \mathcal{X}$ that is not a motion of $\mathcal{X}$.

3. Note that the following quantity

$$\tilde{\measuredangle} ABC = \left[\begin{array}{lll} \pi & \text{if} & \measuredangle ABC = \pi, \\ -\measuredangle ABC & \text{if} & \measuredangle ABC < \pi. \end{array}\right.$$

can serve as the angle measure; that is, the axioms hold if one changes everywhere $\measuredangle$ to $\tilde{\measuredangle}$.

Show that $\measuredangle$ and $\tilde{\measuredangle}$ are the only possible angle measures on the plane, but without Axiom IIIc, this is not longer true.